

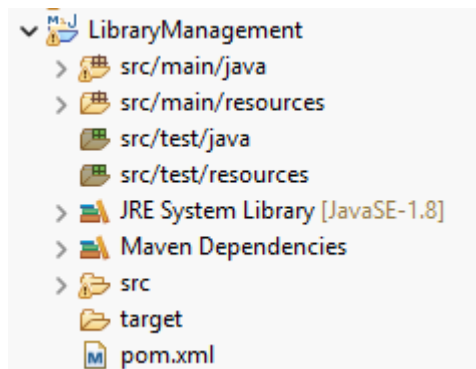
Exercise 1: Configuring a Basic Spring Application

Scenario:

Your company is developing a web application for managing a library. You need to use the Spring Framework to handle the backend operations.

SOLUTION:

Step 1) Set Up a Spring Project



pom.xml:

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <dependencies>
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.1.8</version>
    </dependency>
  </dependencies>
</project>
```

Step 2) Configure the Application Context

applicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="
    http://www.springframework.org/schema/beans
    https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="bookRepository"
    class="com.library.repository.BookRepository"/>

  <bean id="bookService"
    class="com.library.service.BookService">
```

```
        <property name="bookRepository"
            ref="bookRepository"/>

    </bean>

</beans>
```

Step 3) Define Service and Repository Classes

BookRepository.java:

```
package com.library.repository;

public class BookRepository {

    public void displayRepository() {
        System.out.println("BookRepository Bean Created");
    }
}
```

BookService.java:

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
    }

    public void displayService() {
        System.out.println("BookService Bean Created");
        bookRepository.displayRepository();
    }
}
```

MainApp.java:

```
package com.library;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.library.service.BookService;

public class MainApp {

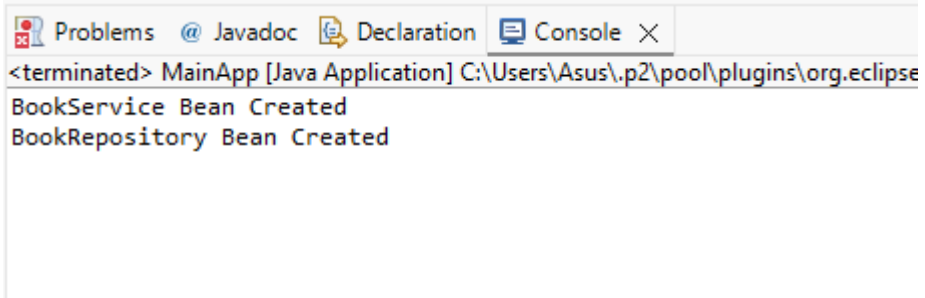
    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            (BookService) context.getBean("bookService");

        service.displayService();
    }
}
```

Step 4) Output



The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console output displays the application's termination and the successful creation of two beans.

```
<terminated> MainApp [Java Application] C:\Users\Asus\.p2\pool\plugins\org.eclipse  
BookService Bean Created  
BookRepository Bean Created
```